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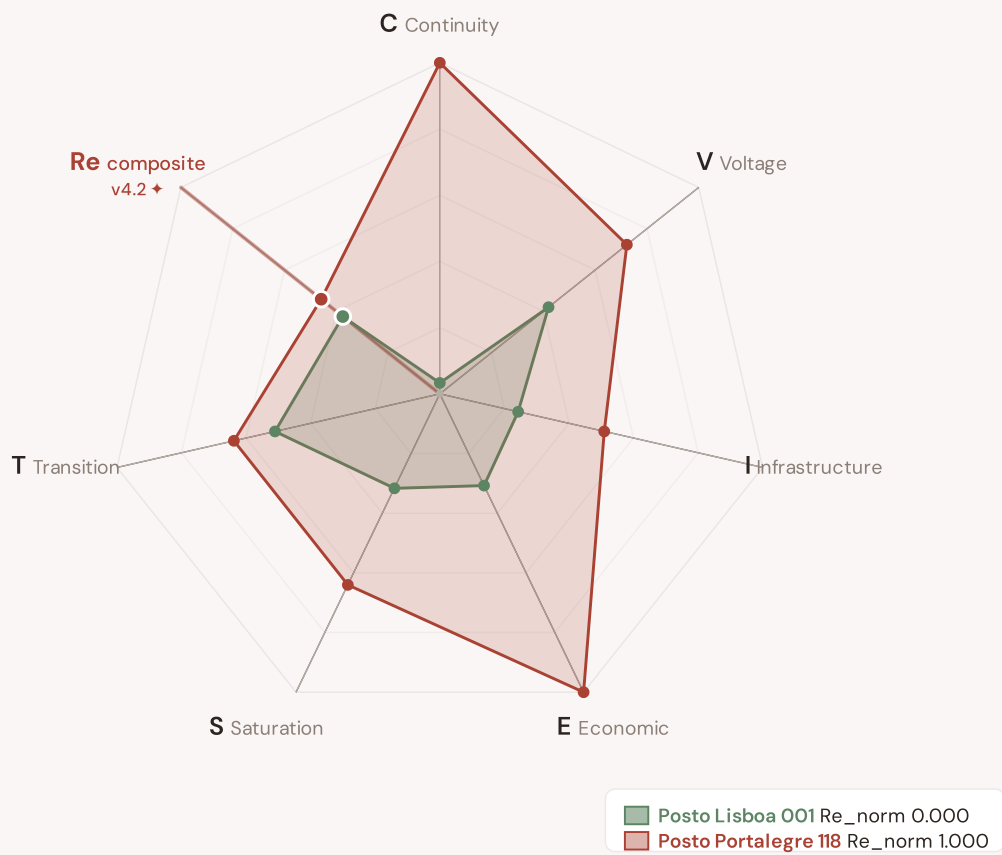
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SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)). Two worked examples overlaid: **Posto Lisboa 001** (low-exposure reference, Re_norm 0.000) · **Posto Portalegre 118** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

10,191

122 EHV · 47 HV · 10,022
distribution-tier

GRID LINES MAPPED

11,043

~18,590 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 10,191 substations and 11,043 grid lines spanning ~18,590 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Portugal's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Posto Porto 2709

Porto, Norte · 60 kV

0.432

Medium

0.140

47th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.2608 · Re_norm 0.176

LOW EXPOSURE

COMPONENTS

C

Continuity: 0.164 / 0.30 (55%)

I

Infrastructure: 0.537 / 0.25 (215%)

S

Saturation: 0.000 / 0.20 (0%)

V

Voltage: 0.846 / 0.10 (846%)

E

Economic: 0.593 / 0.10 (593%)

T

Transition: 0.004 / 0.05 (8%)

MODIFIERS

R3

Consequence: 1.084 ↑ amplifies

R4

Graph Criticality: 0.950 ↓ dampens

R6a

Restoration: 0.930 ↓ dampens

R6b

Seismic: 1.033 ↑ amplifies

R7

Digital Readiness: 0.990 ↓ dampens

This month's spotlight — automatically selected for July 2026 — is **Posto Porto 2709** in Porto, Norte. With an R_final of 0.432 (Medium band, 47th fleet percentile), its risk profile is dominated by the V (Voltage) component at 846% of its weight ceiling, followed by E (Economic) at 593%. Modifiers collectively shift R_base by 31.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

2099

High/Critical + R7 < median

AVG R7 MODIFIER

0.9999

Range [0.99, 1.05]

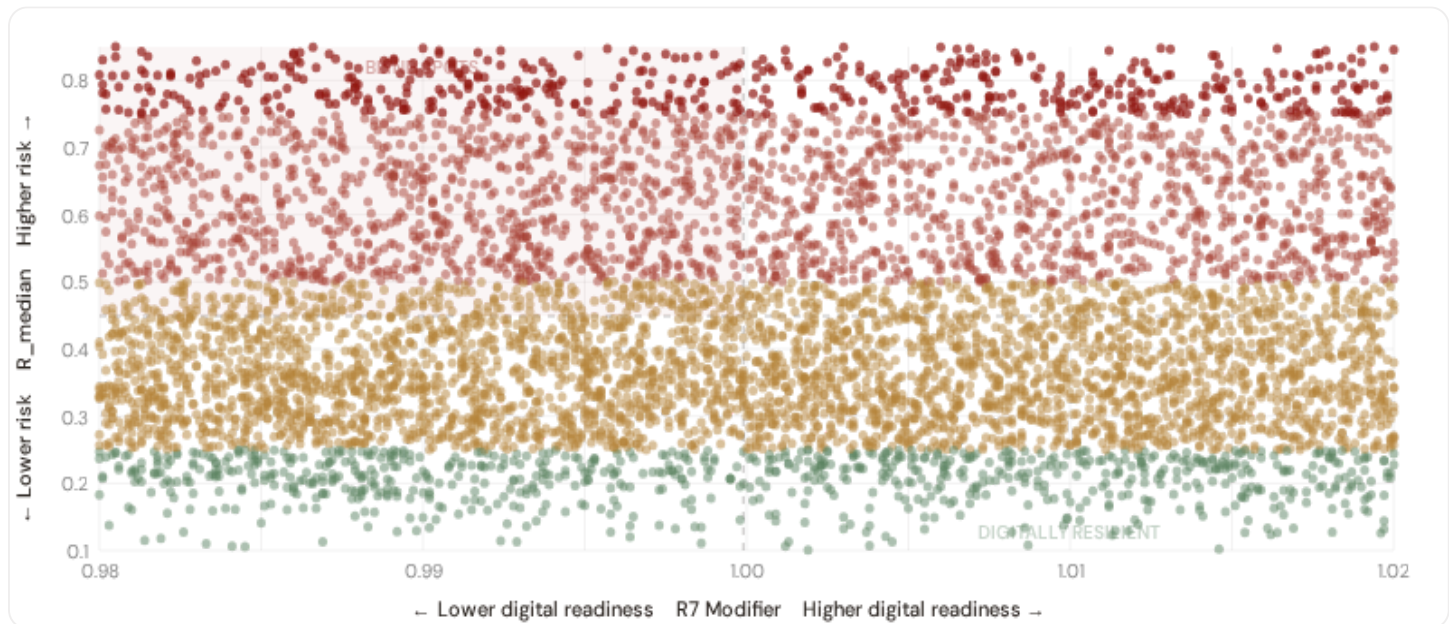
HIGH-CONSEQUENCE SUBS

2,502

24.9% · R3 ≥ 1.07

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



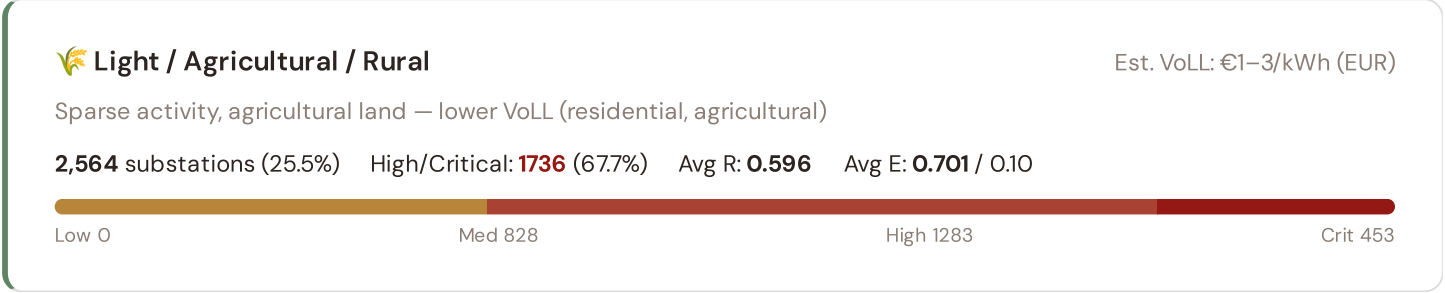
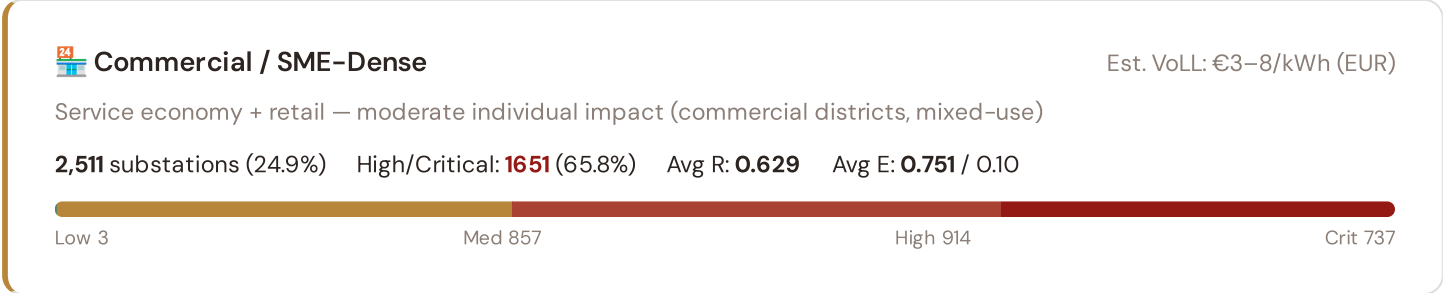
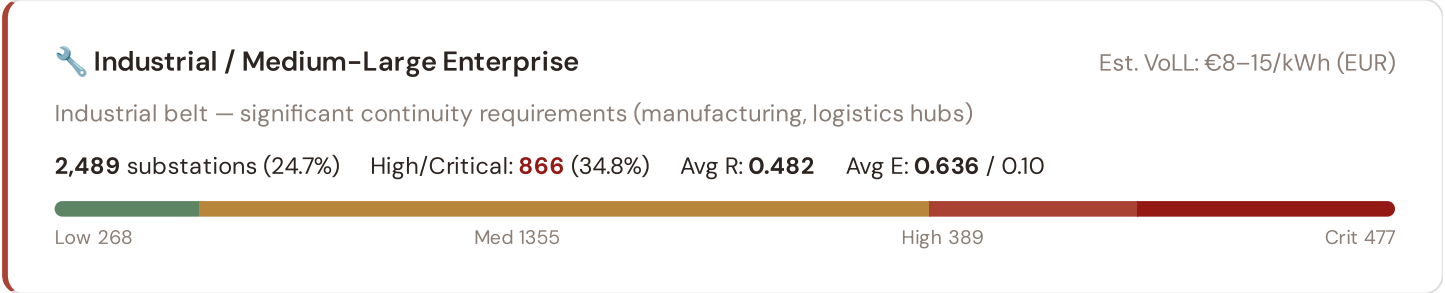
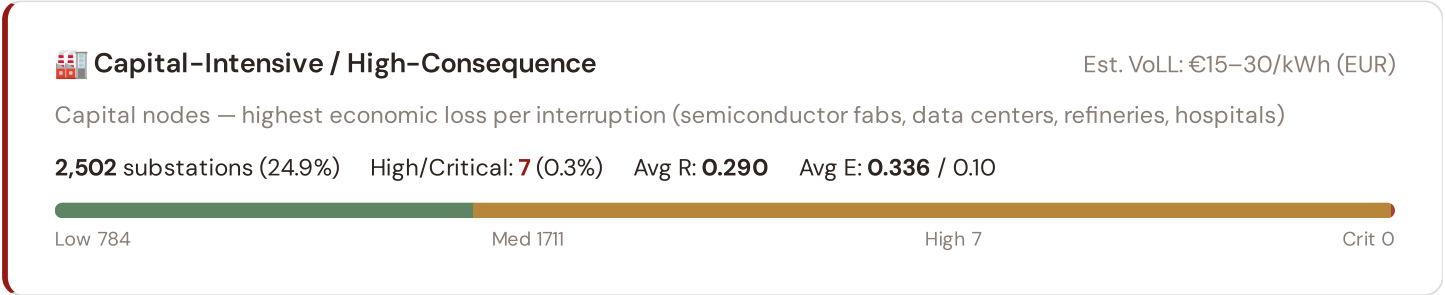
● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **2099 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.9999$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Centro (1396)**, **Norte (281)**, **Alentejo (200)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **7 substations** combine capital-intensive economic fabric (R3 ≥ 1.07) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Norte (1627)**, **Área Metropolitana de Lisboa (875)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 15.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Beja** has the lowest average R7 (0.9990), while **Açores** leads at 1.0027 — a spread of 0.0036 across the R7 range. **9 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

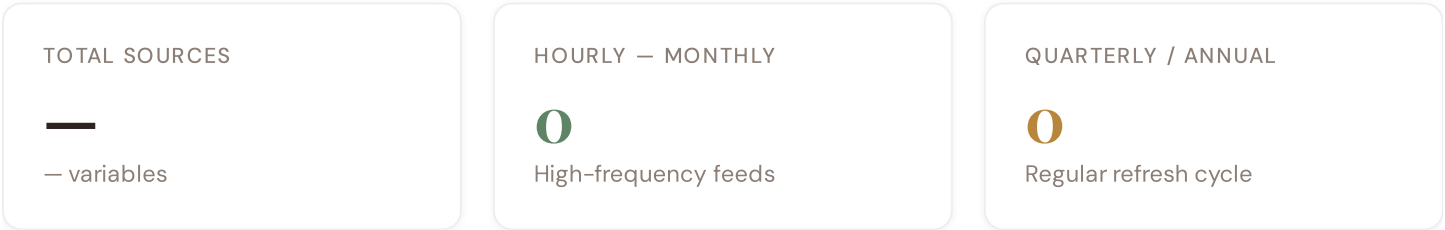
Inaugural edition — Baseline Snapshot (July 2026).

This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9999, median = 0.9999; 0 substations classified HIGH cyber-exposure; 2099 blind-spot substations (High/Critical risk + below-median R7); 803 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



STATIC / DERIVED



Standards + scenarios

Data Source Registry — July 2026

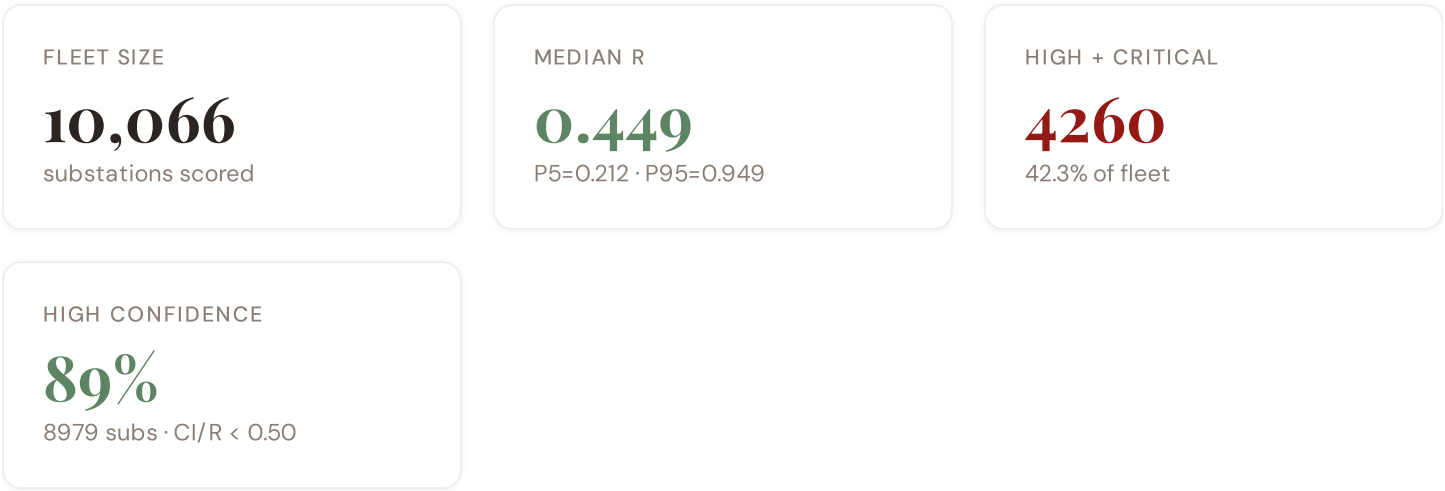
OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
OMIE	OMIE (Operador del Mercado Ibérico de Energía)	MONTHLY	MIBEL zone	2 vars
DGEG	DGEG (Direção-Geral de Energia e Geologia)	QUARTERLY	Concelho	5 vars
MOBI . E	MOBI.E (Rede de Mobilidade Elétrica)	QUARTERLY	Concelho	1 var
ERSE	ERSE (Entidade Reguladora dos Serviços Energéticos)	ANNUAL	Distrito	8 vars
INE	INE (Instituto Nacional de Estatística)	ANNUAL	Distrito	8 vars
ADENE	ADENE (Agência para a Energia)	ANNUAL	Distrito	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
IPMA	IPMA (Instituto Português do Mar e da Atmosfera)	ANNUAL	Distrito	2 vars
APA	APA (Agência Portuguesa do Ambiente)	ANNUAL	Distrito	1 var
AT	AT (Autoridade Tributária e Aduaneira)	ANNUAL	Distrito	1 var
ICNF	ICNF (Instituto da Conservação da Natureza)	ANNUAL	Distrito	1 var
DGEEC	DGEEC (Direção-Geral de Estatísticas)	ANNUAL	Distrito	2 vars
E-REDES	E-REDES Open Data	ANNUAL	Distrito	2 vars
CCDR	CCDR (Comissões de Coordenação Regional)	ANNUAL	Distrito	2 vars
LNEG	LNEG (Laboratório Nacional de Energia e Geologia)	STATIC	Distrito	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars

ISO-9223	ISO 9223 Corrosion	DERIVED	Distrito	1 var
REN	REN (Redes Energéticas Nacionais)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
IPMA-W	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-0M	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **10,066 substation scores remain unchanged** this edition. The fleet median R of 0.449 (mean 0.500) reflects a distribution skewed toward the lower bands, with 10.5% in Low and 47.2% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgoquin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourguoin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	§5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	§5
V42-7	NEW	Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	§7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1-W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 10,191 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

418

15 EHV · 403 HV

HIGH BAND

64 + 354

100.0% of corridor

CORRIDOR MEDIAN R

0.915

Fleet median: 0.449

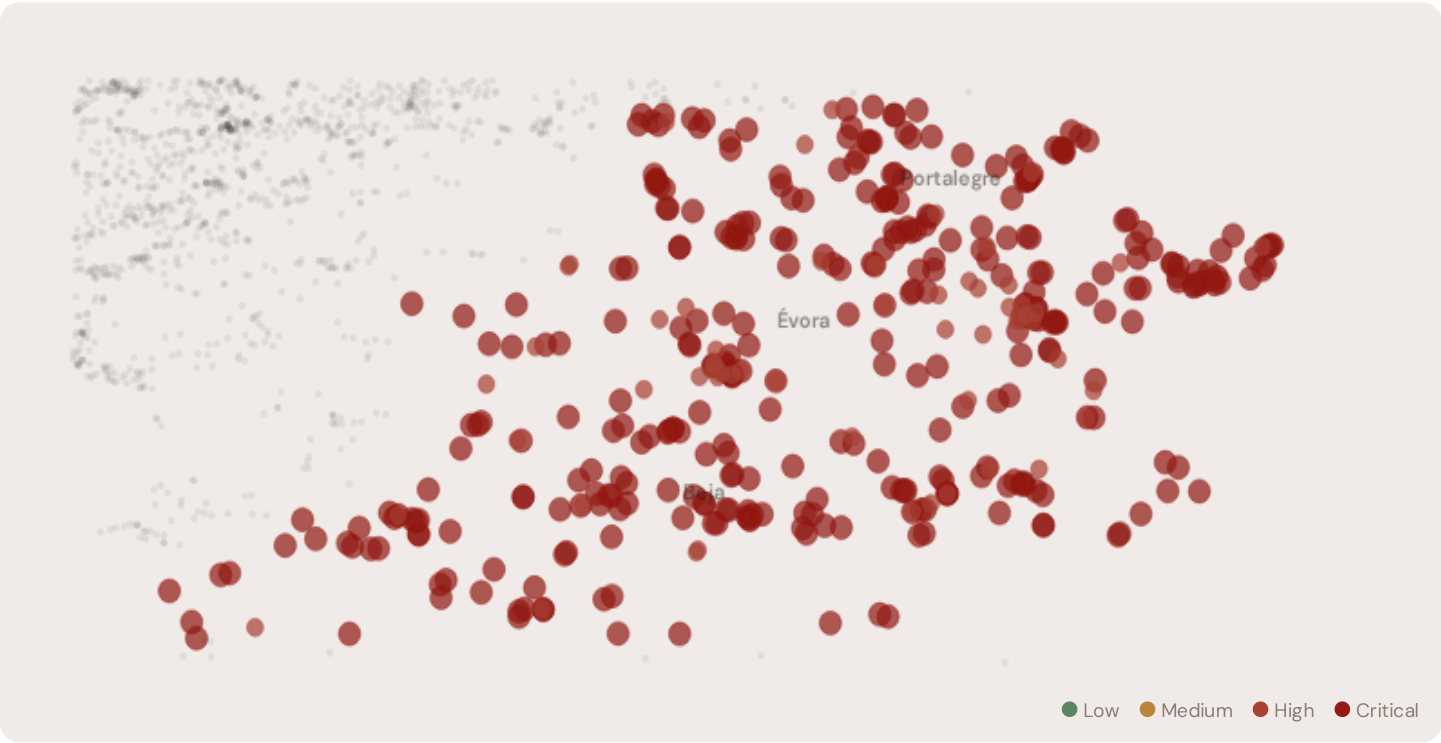
WORST PROVINCE

Portalegre

Avg R: 0.920 · 155 substations

The Alentejo hosts **418 substations** across 3 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **64 substations (15.3%)** are classified High and **354 Critical**, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.449. The corridor median R of 0.915 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
Subestação de Neves-Corvo	Beja	1.050	Critical	1.000	0.718	1.000	0.963	0.998	1.000		
Posto Portalegre 5069	Portalegre	1.050	Critical	1.000	0.623	1.000	1.000	1.000	1.000		
Posto Portalegre 4045	Portalegre	1.049	Critical	1.000	0.873	0.918	1.000	1.000	1.000		

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Posto Portalegre 5999	Portalegre	1.049	Critical	1.000	0.699	1.000	1.000	0.871	1.000		
Posto Beja 1802	Beja	1.049	Critical	0.986	0.879	1.000	1.000	1.000	1.000		
Posto Beja 4739	Beja	1.049	Critical	0.996	0.962	1.000	0.843	1.000	1.000		
Subestação de Tração de Luzianes	Beja	1.048	Critical	0.869	0.908	1.000	0.801	1.000	1.000		
Subestação da Central Fotovoltaica de Tendeiros	Portalegre	1.048	Critical	1.000	0.979	0.840	0.875	1.000	0.734		
Subestação da Central Fotovoltaica de Moura	Beja	1.047	Critical	1.000	1.000	1.000	0.754	1.000	1.000		
Posto Beja 4657	Beja	1.047	Critical	1.000	0.751	1.000	0.837	1.000	1.000		
... 408 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — ALENTEJO VS FLEET			MODIFIER AVERAGES — ALENTEJO VS FLEET		
C — Continuity	0.958 vs 0.355 (+170%)		R3 Consequence	1.004	fleet 1.040 →
I — Infrastructure	0.971 vs 0.298 (+226%)		R4 Graph Crit.	0.958	fleet 0.955 ↓
S — Saturation	0.972 vs 0.400 (+143%)		R6a Restoration	1.062	fleet 0.959 ↑
V — Voltage	0.488 vs 0.502 (−3%)		R6b Seismic	1.043	fleet 1.053 ↑
E — Economic	0.881 vs 0.607 (+45%)		R7 Digital Read.	1.000	fleet 1.000 →
T — Transition	0.955 vs 0.438 (+118%)				

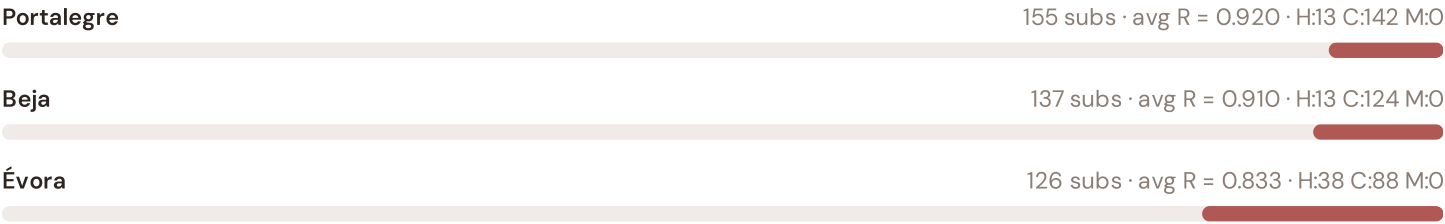
The dominant risk driver across the Alentejo corridor is **T (Transition)**, averaging 0.955 against a fleet average of 0.438 — occupying 1909% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.881 vs fleet 0.607 (881% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.004 (fleet: 1.040), reflecting the socio-economic

exposure of the corridor. The R6b seismic modifier averages 1.043, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The region regions-level breakdown reveals significant intra-regional variation. **Portalegre** leads with a mean R of 0.920 across 155 substations, while **Évora** scores 0.833 — a spread of 0.086 within a single corridor. The SSI's substation-level granularity reveals that the Alentejo is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

412 Alentejo substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Portalegre sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

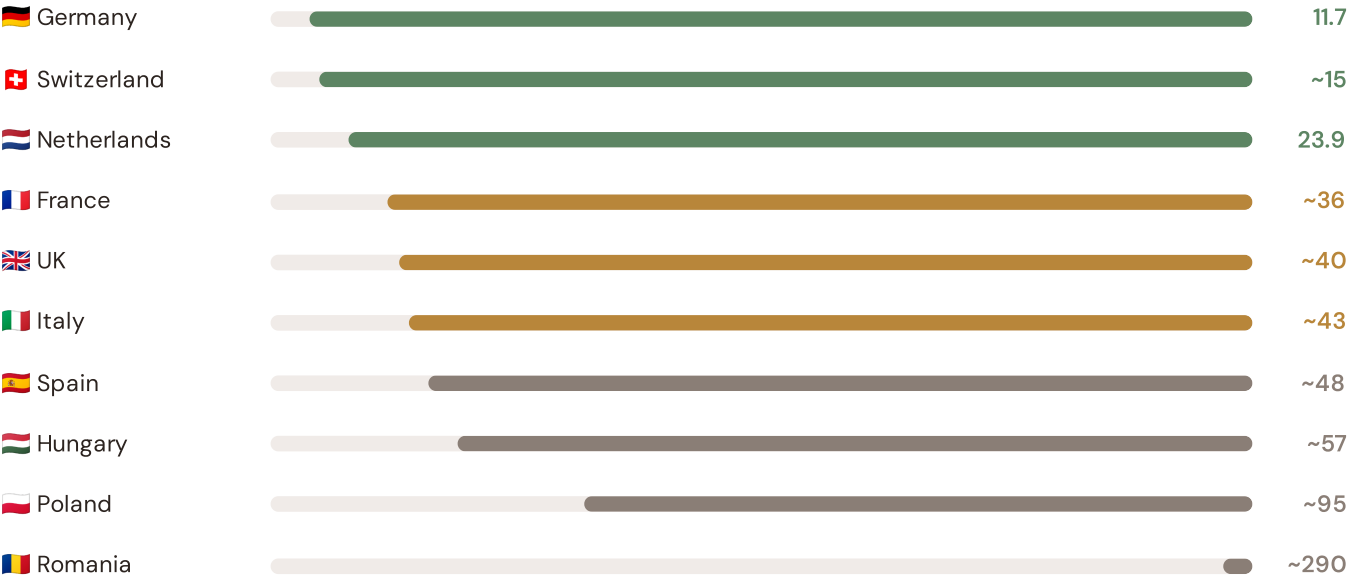
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Portugal's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



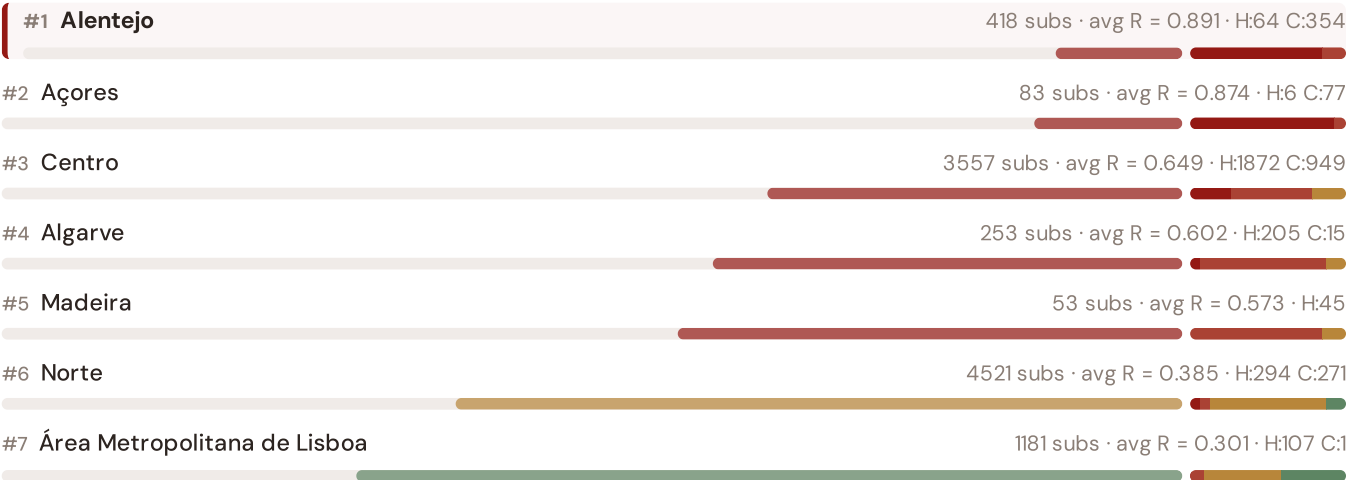
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Alentejo corridor deep-dive shows **418 substations** (of 418) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.958 is **2.7× the fleet average** of 0.355. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regions Risk Ranking — All 7 Portuguese Regions

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Alentejo ranks **#1 of 7 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Alentejo, Açores, Centro, while the three most resilient are Área Metropolitana de Lisboa, Norte, Madeira. 5 of 7 score above the fleet average of 0.500. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

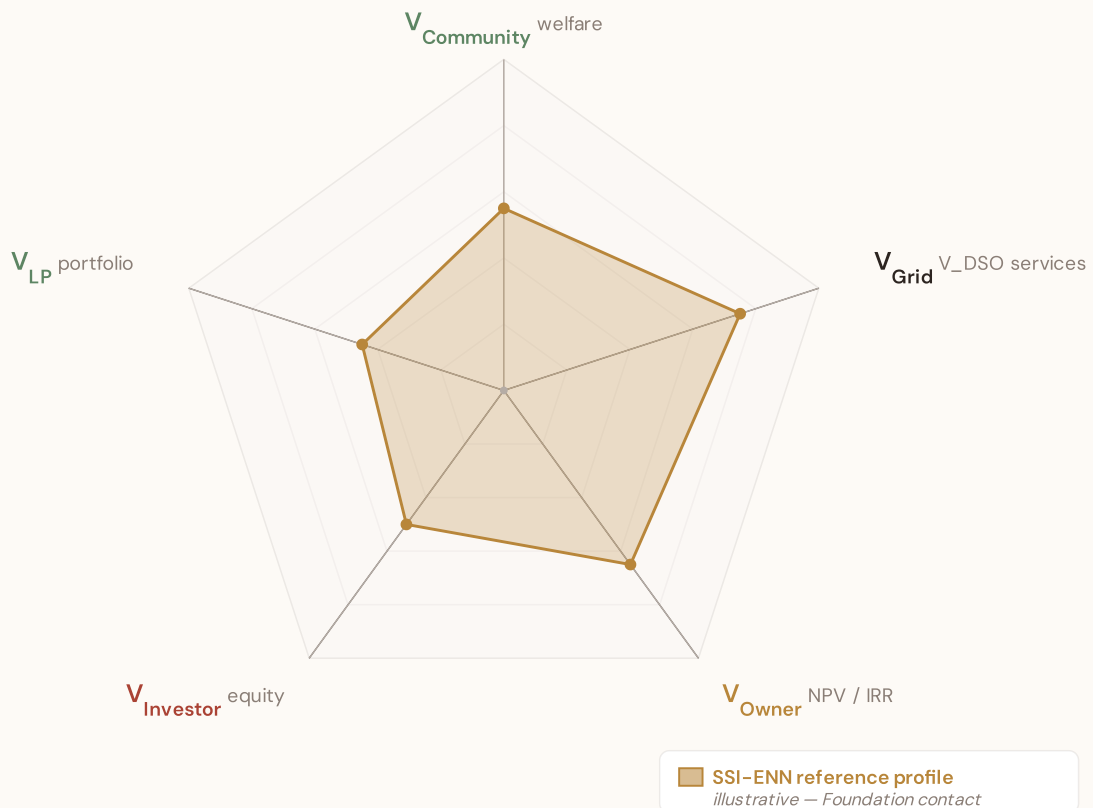
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): $V_{\text{Community}}$ (welfare avoided outages, energy-poverty relief, employment) · V_{Grid} (V_{DSO} — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · V_{Owner} (NPV / IRR / payback per asset class) · V_{Investor} (risk-adjusted equity return at the financing tier) · V_{LP} (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{\text{Community}}$ + V_{Owner} are co-secondary. V_{Investor} + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3

§0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal

prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

→ Quantile regression (pinball loss) + conformal prediction

→ Calibrated against SSI Monte Carlo ground truth

→ Per-country, per-voltage, per-band calibration verification

→ Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 10,066 substations with complete feature vectors. Average CI_width of 0.183 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 1055 Low, 4751 Medium, 2593 High, 1667 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 2

 Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill W1 · W2 · W3 · W4 · W8 <i>Peer-reviewed methodology operational at fleet scale.</i>	Published-baseline (1) ▲ — solid fill + chevron corner marker W9 — Supply Security <i>Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).</i>	Baseline-only (1) — striped fill W5 — Digital Inclusion <i>Foundation methodology under development; baseline value defensible at v4.2.</i>
Commercial (3) — dashed-faded W6 · W7 · W10 <i>Commercial deliverables — please contact the Foundation.</i>		

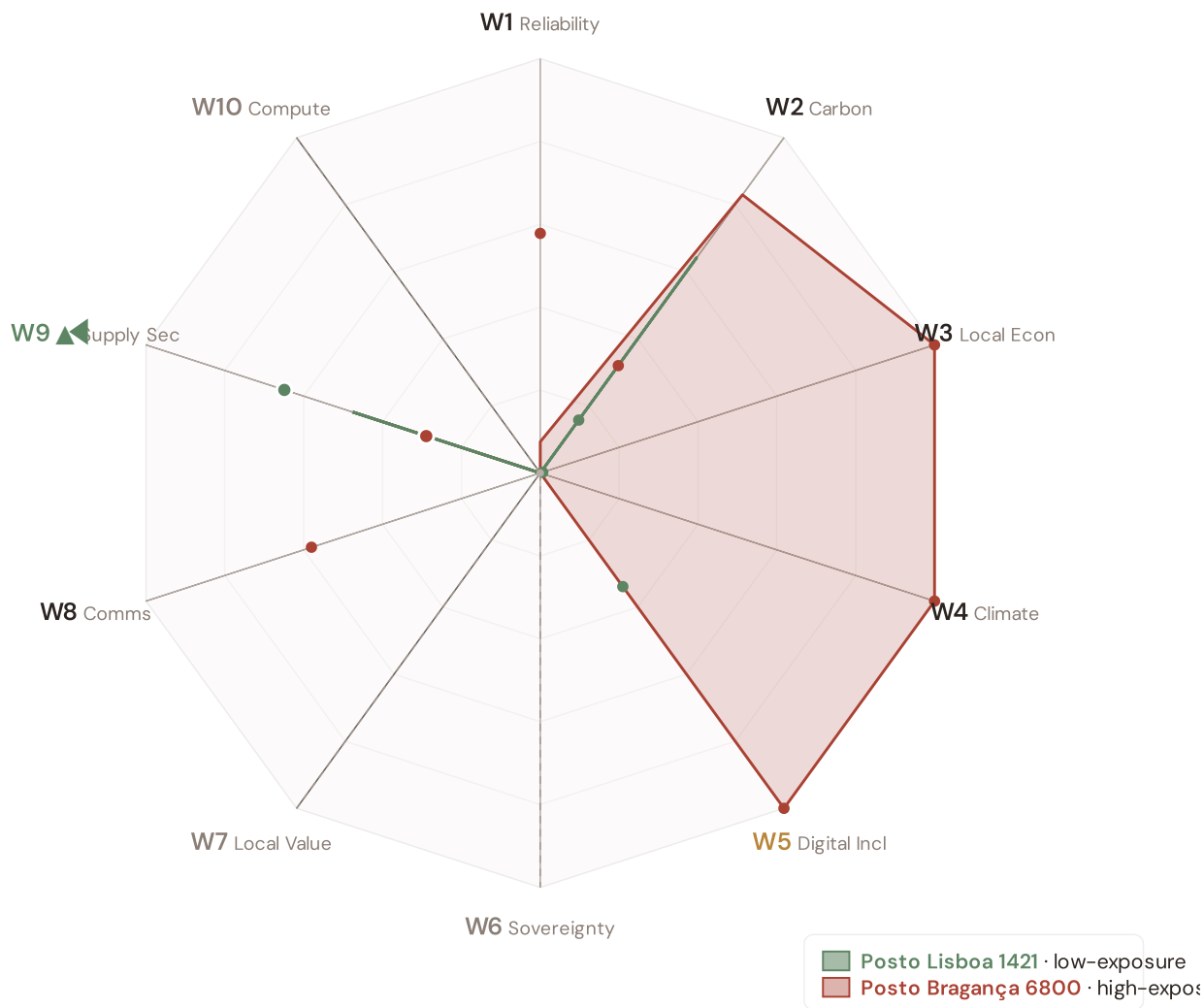
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Portugal, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Posto Lisboa 001** (low-exposure reference, Re_norm 0.000) and **Posto Portalegre 118** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Posto Lisboa 1421 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Posto Bragança 6800 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (\text{SAIDI} + + \text{-POV})$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ \text{EEA})$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$ resilience modifier family
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM} + + \text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population} (\text{Markov 10-yr critical probability})$
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) -$ OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature.

Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — Posto Lisboa 1421 (low-exposure reference) and Posto Bragança 6800 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6

§2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Capital Metropolitan

Deep-dive into Region 02's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Capital-intensive tier + flood overlay.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ERSE (Entidade Reguladora dos Serviços Energéticos)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

770 Bridge Substations — Single Points of Failure

770 substations are topological bridges — their failure disconnects part of the network. Of these, **417** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Region 02**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Portuguese utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 10,191 substations · 11,043 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.